Understanding Negative User Responses to Generative AI Errors in E-Business Information Systems through Expectation Rigidity and Cognitive Shock*

Hyeri Yang, Vireak Yeu, and Hyunchul Ahn†
Graduate School of Business IT, Kookmin University, Republic of Korea
{imhyeri, v020610, hcahn}@kookmin.ac.kr

Abstract

Generative AI is increasingly embedded in e-business information systems, but its erroneous outputs may trigger negative user reactions. This study proposes a research model explaining how users develop anger and dissatisfaction when generative AI fails. Drawing on expectation-confirmation theory and the violation of expectation perspective, the model suggests that AI dependence and anthropomorphism increase expectation rigidity, which intensifies cognitive shock when users encounter AI errors. Cognitive shock, in turn, is expected to increase anger and dissatisfaction. Error severity is also proposed as a moderator strengthening these relationships. This research contributes to e-business information systems literature by shifting attention from AI adoption to the cognitive and emotional consequences of AI service failure.

1 Introduction

Generative AI services based on large language models (LLMs) are rapidly becoming part of e-business information systems, supporting tasks such as information search, customer support, document generation, and decision support. However, these systems may also produce

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† Corresponding author

hallucinations, fabricated references, or misleading outputs (Ji et al., 2023). Such failures are important because users may rely on AI-generated outputs in business-related contexts.

Prior information systems research has mainly examined adoption, continuance, trust, and satisfaction (e.g. Bhattacherjee, 2001). In contrast, less is known about how users emotionally respond when AI-enabled systems fail. To address this gap, this study develops a research model explaining how generative AI errors lead to negative user responses through expectation rigidity and cognitive shock.

2 Research Model and Hypotheses

The model suggests that users' negative responses to generative AI errors depend not only on the error itself but also on prior expectations. AI dependence and anthropomorphism are expected to increase expectation rigidity, as repeated reliance and human-like interaction may lead users to view generative AI as reliable and socially responsive (Bhattacherjee, 2001; Parasuraman & Manzey, 2010; Nass & Moon, 2000). When such rigid expectations are violated by AI errors, users may experience cognitive shock, which in turn increases anger and dissatisfaction. Error severity is expected to further strengthen these effects (Bougie et al., 2003).

Accordingly, this study proposes the following hypotheses:

H1. AI dependence positively influences expectation rigidity.

H2. Anthropomorphism positively influences expectation rigidity.

H3. Expectation rigidity positively influences cognitive shock.

H4a. Cognitive shock positively influences anger.

H4b. Cognitive shock positively influences dissatisfaction.

H5a. Error severity strengthens the relationship between cognitive shock and anger.

H5b. Error severity strengthens the relationship between cognitive shock and dissatisfaction.

3 Expected Contributions and Conclusion

This study contributes to e-business information systems research by shifting attention from AI adoption and satisfaction to the negative emotional consequences of AI service failures, introducing expectation rigidity and cognitive shock as explanatory mechanisms, and distinguishing anger from dissatisfaction as separate user responses. Practically, it suggests that AI-enabled e-business system providers should manage not only technical accuracy but also user expectations through uncertainty signaling, verification reminders, transparent error explanations, and recovery strategies, thereby offering a foundation for future empirical research on AI error experiences and expectation management.

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